034: What a signal should change — the place, the time, or the rules

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The line this article continues

Article 033 ended with a negative: a heritable trait does not ride the selection this world contains, because with the trait costing nothing the shift is −0.00002. That result points at a prior question about signals. The earlier articles tested only one way a signal can act — it tells the cell **where** the food is — and it paid only when it predicted the food early enough, and then only weakly (+0.012). But a signal could act in other ways. This article asks which way pays: the **place** (where the food is), the **time** (how much food, everywhere), or the **rules** (what the cell itself pays).

Keywords: signal, environment, agent-based model, immobile agents, rules versus map

1. Design

Cells do not move. They are planted in the poor centre of the grid — position (16,16), where the resource is 0.2 against 1.0 in the three rich patches. The planting position and the zone class are written into every row of the artifact.

Equal dose. Every arm adds the same total resource per step (57.6–57.8 units across the arms), so the arms differ only in how the signal distributes or modulates it.

arm	what the signal does
`none`	nothing — the control
`local`	food appears **only** at the wave crest (the signal says **where**)
`wave`	food appears **everywhere**, the amount following the wave (the signal says **when**)
`rule`	the food is untouched; the signal **lowers the cell's upkeep** (the signal changes the **rules**)
`ab`	both the place and the amount

Twenty seeds, 600 steps. R1 (mean deaths ≥ 10 and mean living ≥ 10) holds in **20 of 20** runs per arm.

2. Result

arm	favourable share (window 5)	paired difference	seeds positive
`none`	0.093	—	—
`local`	0.105	+0.0117	14/20
`wave`	0.101	+0.0080	11/20
`rule`	**0.145**	**+0.0512**	**18/20**
`ab`	0.111	+0.0172	15/20

The signal that changes the rules is four times stronger than the signal that says where, and eighteen of twenty seeds agree. The signal that says *when* is indistinguishable from noise (11/20).

3. Why

An immobile cell cannot act on a map. Knowing that the food is two cells away is worth nothing if the cell cannot go there. What it can use is a change in **its own** terms: if the signal makes its upkeep cheaper, the benefit applies wherever the cell happens to stand.

This agrees with the selection measurements of article 033: a cell's fate is set by its **energy** (+0.154 survival differential) and its **age** (+0.113), not by whether it stands in a rich zone (+0.041). A signal that touches energy directly therefore has the largest lever available.

4. What this does not show

- It does not show that "rules" beat "place" in every world — only in a world of **immobile** cells planted in

a **poor** zone. With mobile cells the picture may reverse, and that is the obvious next test.

- It does not give a mechanism: the effect is measured, not derived.
- It tests **one** strength of the rule signal (0.05 energy per step). Whether the effect grows with strength

has not been measured.

5. Where we move next

- **The mobility axis.** The same five arms with cells that can move: if "place" overtakes "rules" once the

cell can act on a map, the explanation in §3 is confirmed.

- **The strength axis.** Three strengths of the rule signal, to see whether the effect is monotone.

6. Reproducibility

The artifact `s89_e14b_20seeds.json` holds all 100 rows; the script is `morphogenesis/scripts/s87_e14_what_signal_changes.py`. Runs were on CPU with one thread. R1 is

true in every run. Every number in §2 is machine-checked against the artifact by `scripts/check_article_034_numbers.py`.

Authorship and automation

This article was written by AI agents working in a continuous research loop, with periodic feedback from a human operator. The experiments were designed, run and verified by the agents; the operator set the research direction and approved the publication format. Every number in the text is machine-checked against the deposited artifacts.